EXP-18 : To verify that each subkey takes 24 bits from each disjoint 28-bit half.

PROGRAM :

Run | Debug | Stop | Share | Save | { } Beautify

main.c

```c
#include <stdio.h>
int pc1[]={57,49,41,33,25,17,9,1,58,50,42,34,26,18,
 10,2,59,51,43,35,27,19,11,3,60,52,44,36,63,55,
 47,39,31,23,15,7,62,54,46,38,30,22,14,6,61,53,
 45,37,29,21,13,5,28,20,12,4};
int pc2[]={14,17,11,24,1,5,3,28,15,6,21,10,23,19,12,4,
 26,8,16,7,27,20,13,2,41,52,31,37,47,55,30,40,
 51,45,33,48,44,49,39,56,34,53,46,42,50,36,29,32};
int main(void) {
 int disjoint=1, left=0, right=0;
 for (int i=0;i<28;i++) for (int j=28;j<56;j++)
 if (pc1[i]==pc1[j]) disjoint=0;
 for (int i=0;i<24;i++)
 if (pc2[i]>=1 && pc2[i]<=28) left++;
 for (int i=24;i<48;i++)
 if (pc2[i]>=29 && pc2[i]<=56) right++;
 printf("C0 and D0: 28 bits each\n");
 printf("Disjoint original-key sets: %s\n",
 disjoint ? "Yes":"No");
 printf("First 24 subkey bits from C: %d\n",left);
 printf("Last 24 subkey bits from D: %d\n",right);
 puts("Independent rotations preserve this in every round.");
 return 0;
}
```

OUTPUT :

```
C0 and D0: 28 bits each
Disjoint original-key sets: Yes
First 24 subkey bits from C: 24
Last 24 subkey bits from D: 24
Independent rotations preserve this in every round.


...Program finished with exit code 0
Press ENTER to exit console.
```